Mohamed Khaled Rammah *Machine Learning Engineer*

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Education

BS **Assiut National University**, Computer Science and Artificial Intelligence — Oct 2022 – July 2026

- GPA: 3.9/4.0 (Academic Record)
- **Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Design Patterns, Operating Systems, Deep Learning, Probability and Statistics, Linear Algebra, Computer Vision, Natural Language Processing, Database Systems, Computer Networks

Activities

- **Competitive Programming:** Placed 1st in the ECPC-2023 Qualifications (university level), 83rd in ECPC-2023, and 76th in ECPC-2024 Qualifications .
- **Problem Solving:** Solved +120 problems at LeetCode | Solved +1100 problems at Codeforces
- **Events:** Attended several university conferences and tech events focused on Artificial Intelligence and emerging technologies, including *Google Developer Groups On Campus – Assiut University*.

Projects

Flower Image Classification using CNNs — Flower Classification CNN

- Developed and trained convolutional neural networks (CNNs) to classify images of flowers into multiple categories using the TensorFlow Flowers dataset.
- Implemented preprocessing pipelines, including normalization, caching, shuffling, and prefetching, to optimize training performance.
- Built an enhanced CNN with data augmentation and dropout layers, achieving improved generalization and robust accuracy on validation data.
- Visualized model performance with accuracy/loss curves and confusion matrices, and deployed predictions on new unseen flower images.
- Tools & Stack: Python, TensorFlow, Keras, NumPy, Matplotlib, Seaborn

Cancer Detection Web Application — Cancer Detection ML App

- Developed a machine learning web application to predict cancer presence based on patient data and gene expression patterns.
- Implemented and compared multiple ML models, including K-Nearest Neighbors (KNN) and Naive Bayes, achieving accurate and efficient predictions.
- Integrated feature scaling (StandardScaler) and data preprocessing to improve model performance and precision.
- Designed a simple, responsive web interface for patient data input and result interpretation.
- Tools & Stack: Python, Flask, Scikit-learn, HTML5, CSS3, Git, GitHub

Internships

Machine Learning Internship – Information Technology Institute (ITI) **ITI Repo** — July 2025 – Aug 2025

- Completed an intensive 2-week internship focused on Artificial Intelligence and Machine Learning.
- Covered topics including: Introduction to AI, Python for Data Science, NumPy, Data Preparation and Exploration.

- Gained strong foundations in Linear Algebra, Probability, and Statistics for ML.
- Practiced building and training ML models, including Neural Networks and Deep Learning with practical labs.

SKILLS

Programming languages: C++, C, Java, C# , SQL, JavaScript, Python

ML & AI Frameworks: Scikit-learn, TensorFlow, PyTorch, Keras.

Data Science Tools: NumPy, Pandas, Matplotlib, Seaborn.

Databases: SQL, MongoDB.

Concepts: Machine Learning, Deep Learning, NLP, Computer Vision, Data Analysis.

Development Tools: Git, GitHub.

Soft skills: Problem solving, teamwork, and time management.

Books

- **Grokking Algorithms:** (Book Repo) Learned and practiced fundamental algorithmic concepts including: - Binary Search - Big-O Notation and Complexity Analysis - Recursion and Divide & Conquer - Sorting Algorithms (Selection Sort, Quicksort, Merge Sort) - Hash Tables - Breadth-First Search (BFS) & Depth-First Search (DFS) - Dijkstra's Algorithm (Shortest Path) - Greedy Algorithms - Dynamic Programming (e.g., Knapsack, Longest Common Subsequence) - K-Nearest Neighbors (Intro to ML perspective)
- **The Object-Oriented Thought Process:** Learned and practiced core OOP concepts including: - Classes and Objects - Abstraction - Encapsulation - Inheritance - Polymorphism - Interfaces and Abstract Classes - Composition vs. Inheritance - Object Relationships (Association, Aggregation, Composition) - Coupling and Cohesion - Design Principles and Best Practices
- **Head First Design Patterns:** Studied and applied key software design patterns including: - Strategy Pattern - Observer Pattern - Decorator Pattern - Factory Method & Abstract Factory Patterns - Singleton Pattern - Command Pattern - Adapter Pattern - Facade Pattern - Template Method Pattern - Iterator & Composite Patterns - State Pattern - Proxy Pattern - MVC (Model–View–Controller) Architecture Concepts

Courses

- **The Complete Data Structures and Algorithms Course in Python** – Dr. Mustafa Saad Covered arrays, linked lists, stacks, queues, recursion, binary search, sorting, merge sort, complexity analysis, and OOP.
- **Design Patterns** – University of Alberta via Coursera Studied reusable and scalable software design patterns.
- **Complete Web Development Course** – Hitesh Choudhary Comprehensive training on HTML, CSS, Tailwind CSS, Node.js, React, MongoDB, Prisma, and deployment techniques.
- **Supervised Machine Learning: Regression and Classification** – DeepLearning.AI via Coursera Studied regression and classification techniques for supervised machine learning.
- **Cloud and Virtualization Concepts** – MaharaTech – ITI/Mooc Explored cloud computing models (IaaS, PaaS, SaaS) and virtualization technologies (VMware, VirtualBox, Docker).